Congratulations for completing course!

- Congratulations for completing this course!
- This week, you learned how to estimate unknown parameter values of a state-space model using xKF if states are known.
- Then, you learned how to estimate unknown states and parameter values simultaneously using xKF and both joint and dual-estimation approaches.
- You learned how to implement the parameter-estimation and joint-estimation methods in Octave and saw some results from applying them to simple problems.
- This brings us to the end of course 3 on nonlinear Kalman filters.

Where to from here?

- Course 4 explores particle filters for estimating the state of systems having very nonlinear dynamic models.
- You will learn:
  - Background concepts relating to the goal of particle filters—to estimate the conditional pdf $f(x_k \mid \mathbb{Z}_k)$ accurately and efficiently.
  - A brute-force method to estimate $f(x_k \mid \mathbb{Z}_k)$ that is computationally too complex for most applications.
  - How to estimate integrals efficiently (via Monte–Carlo method) as a step toward simplifying the computation of $f(x_k \mid \mathbb{Z}_k)$.
  - How to derive the particle filter using Monte–Carlo integration as its basis, and some significant problems with this approach.
  - How to correct these problems to derive a robust implementation of particle filters.
- I look forward to sharing these skills with you!